Pediatric Wide-Complex Tachycardia

PC-06 | Pediatric Cardiac

Quick flow

01 Wide-Complex Tachycardia [EMT / AEMT / Paramedic]

Assess perfusion, mental status, BP, respiratory status, QRS pattern, cardiac history, and reversible causes.

02 Cardiopulmonary Compromise? [EMT / AEMT / Paramedic]

Hypotension • acutely altered mental status • signs of shock • sudden collapse. If yes, prepare synchronized cardioversion immediately.

03 Synchronized Cardioversion [Paramedic]

0.5-1 J/kg; if unsuccessful, increase to 2 J/kg. Confirm sync mode before every shock. Sedate only if immediately available and it does not delay treatment.

04 Stable Rhythm Pattern? [EMT / AEMT / Paramedic]

Obtain early Medical Control/expert consultation and Paramedic rhythm evaluation. Continue monitoring and transport support.

05 Medication Strategy [Paramedic]

For stable regular monomorphic rhythm, adenosine only with Medical Control/expert consultation. All other antiarrhythmic decisions are Medical-Control-directed.

06 Reassess + Transport [EMT / AEMT / Paramedic]

Continuous monitoring, rhythm strips, cause-directed treatment, and early pediatric-capable destination notification.

Overview

- Treat pediatric wide-complex tachycardia as potentially ventricular until proven otherwise. A wide QRS is greater than 0.09 seconds.
- Cardiopulmonary compromise requires urgent synchronized cardioversion; stable medication decisions require early Medical Control/expert consultation.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Support airway and oxygenation as clinically indicated, identify cardiopulmonary compromise, obtain rhythm strip when available, and prepare for synchronized cardioversion.
- Assess reversible causes including hypoxia, toxin/stimulant exposure, fever, cardiac history, and symptoms of poor perfusion.

AEMT

- Perform all EMT care plus IV/IO access and supportive treatment within current standing orders.
- Do not delay cardioversion or transport for repeated vascular-access attempts.

Paramedic

- Perform all prior care plus synchronized cardioversion for cardiopulmonary compromise and early Medical Control/expert consultation for stable wide-complex tachycardia.
- Use adenosine only for a stable, regular, monomorphic wide-complex rhythm and only with Medical Control/expert consultation.

- Document rhythm before and after each intervention; obtain a 12-lead when stable without delaying treatment.

Indications

- Pediatric tachycardia with QRS greater than 0.09 seconds and suspected VT, SVT with aberrancy, or polymorphic tachycardia.

Treatment steps

- Treat a wide-complex rhythm as potentially ventricular until proven otherwise. Obtain early Medical Control/expert consultation before antiarrhythmic treatment in a stable child.
- Do not give adenosine for unstable, irregular, or polymorphic wide-complex tachycardia.
- Assess and treat reversible causes: hypoxia, electrolyte/metabolic abnormality, toxin/stimulant exposure, myocarditis/cardiomyopathy, and congenital heart disease.
- Use continuous ECG, SpO_2, blood pressure, and rhythm strips before and after interventions. Obtain a 12-lead only when it does not delay treatment.
- If the child becomes pulseless, begin CPR and move immediately to PC-04. If the rhythm is VF/pulseless VT, use PC-07.

Medications

Adenosine

0.1 mg/kg rapid IV/IO push followed immediately by rapid saline flush; maximum 6 mg. If needed, repeat 0.2 mg/kg; maximum 12 mg. Only for stable, regular, monomorphic wide-complex tachycardia with Medical Control/expert consultation.

Midazolam

0.1-0.2 mg/kg IV/IO/IN for cardioversion; maximum single dose 2 mg, maximum total 5 mg

- Use only if immediately available; do not delay synchronized cardioversion. Verify against the current Claiborne pediatric medication reference.

Antiarrhythmic therapy

Agent and dose determined by Medical Control; do not administer amiodarone and procainamide together.

Warnings / precautions

- Do not delay synchronized cardioversion for sedation, 12-lead acquisition, or repeated IV attempts.
- Do not give adenosine for unstable, irregular, or polymorphic wide-complex tachycardia.
- Do not administer amiodarone and procainamide together.
- If pulseless at any time, begin CPR and move immediately to Pediatric Pulseless Arrest.

Clinical pearls

- Adenosine can be diagnostic or therapeutic only in the stable, regular, monomorphic wide-complex rhythm; expert consultation is part of that decision.
- Stable pediatric wide-complex tachycardia still warrants early specialty-capable destination planning because myocarditis, cardiomyopathy, toxins, and congenital disease may be causes.

Related pathways / contacts

AR-05 Pediatric Airway - Open for airway and oxygenation support.

PC-04 Pediatric Pulseless Arrest - Open immediately if the child becomes pulseless.

PC-05 Pediatric Narrow-Complex Tachycardia - Open for narrow-complex tachycardia.

PC-07 Pediatric VF / Pulseless VT - Open for shockable pulseless arrest.

Medication Reference - Open the current Claiborne medication reference.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

Revision: 2026-08-14 | Last verified: 2026-08-14